Moein Bagheri

AI product builder with an ear for detail · Speech & Audio AI

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Final-year Computer Engineering student (AI track) and AI-native builder: quick to build, careful to check, down to the details people feel but rarely notice. Focused on speech, audio and language. Available full-time for applied AI roles, especially in NLP, speech and signal processing.

EDUCATION

B.Sc. in Computer Engineering, University of Isfahan — Sep 2022 – Jun 2027

- GPA **17.26/20** overall · **18.5/20** over the last three semesters and in specialized courses · AI major, IT minor
- Selected courses: Signals & Systems (20) · Language & Speech Processing (20) · Design of Algorithms (20) · Data Mining (19.75) · Applied Linear Algebra (18.87) · Deep Learning (18.5)
- Final-year project (in progress): Persian–English code-switched speech recognition · Supervisor: Dr. S. F. Noorbehbahani

NODET (SAMPAD) High School, Yazd · *Iran's schools for exceptional talents* — Diploma

SELECTED PROJECTS

Speech Emotion Recognition — PyTorch · librosa · 2026

- Trained plain and residual 1D CNNs on MFCC features to recognize four emotions in IEMOCAP (4,901 clips, 10 speakers).
- Speaker-independent evaluation showed what a random split hides: accuracy fell from 66.7% to 52.6% on unseen speakers, and "happy" was mostly heard as neutral or sad.

Local Research Assistant (RAG) — Ollama · FAISS · Streamlit · 2026

- Built a local assistant over four retrieval papers (REALM, DPR, RAG, FiD) that cites the page behind each answer.
- Retrieval raised answers containing the reference fact from 8% to 75% on a 12-question test.

Persian Search Engine *(with S. Heydari)* — Python · Hazm · 2026

- Implemented an inverted index, Boolean search, TF-IDF and BM25 by hand over 100 Persian thesis records.

Deep ECG Classification — TensorFlow · 2026

- Trained GRU networks for five-class heartbeat classification on MIT-BIH (93.3% accuracy); per-class analysis showed the headline hid 6% recall on supraventricular beats.

More on GitHub: n-gram melody completion · neural style transfer · YOLOv8 gesture detection · DiaTech product discovery

EXPERIENCE

Teaching Assistant, University of Isfahan — Spring 2026 – Present

Signals & Systems (Dr. M. H. Bateni) · Human-Computer Interaction and E-Commerce (Dr. S. F. Noorbehbahani)

E-Commerce Lab, University of Isfahan — Summer 2025

Business Development Intern · Supervisor: Dr. S. F. Noorbehbahani

- Researched the market for two startup concepts: an online physiotherapy platform and a children's storybook service.
- Took part in the hands-on work of setting up the Computer Innovation Center's workspace ahead of its opening.

LEADERSHIP & HONORS

- Founded and ran Bartarinha, a team math competition between Yazd's NODET boys' and girls' schools (2017–18).
- Hosted the Computer Engineering department's welcome ceremony for new students, at the faculty's invitation.
- Third place in piano, Yazd students' cultural and arts competitions (2019).

SKILLS

Speech & ML	PyTorch, TensorFlow/Keras, scikit-learn, librosa, OpenCV, Ultralytics YOLO
NLP & retrieval	Hazm, NLTK, FAISS, Ollama, retrieval-augmented generation
Code & tools	Python, Java, C/C++, SQL, JavaScript, Git, Streamlit
AI-native dev	Claude Code, Codex, agentic coding workflows
Languages	Persian (native), English (fluent)